



Dolichos falcata Klein attenuated the inflammation induced by monosodium urate crystals *in vivo* and *in vitro*



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ABSTRACT

Ethnopharmacological relevance: *Dolichos falcata* Klein (DF), a Chinese Dai ethnic medicine popularly known as “Tuoyeteng” in Yunnan province of China, has been widely used as a traditional herbal medicine for the treatment of fracture and beriberoid disease for a long time in China. The present study was carried out to investigate the anti-inflammatory activity and the bioactive chemical constituents of DF, and further to assess its possible mechanism on gouty arthritis in an animal model of the MSU crystals-induced gouty inflammation.

Materials and methods: The ethanol extract (EE) of DF at the doses of 10, 20 and 40 mg/kg was administered to the rats treated with MSU crystals to evaluate the anti-gouty arthritis effect. Subsequently, the components of EE were isolated and identified using classical methods. Phyto-chemical analysis of EE was further carried out by HPLC–DAD. Finally, the anti-inflammatory effect of EE and two isolated components were assessed using the MSU crystals-treated monocyte/macrophage cell line RAW 264.7 *in vitro*.

Results: EE (10, 20 and 40 mg/kg) significantly attenuated the pain threshold value, the joint swelling degree, the inflammatory cell infiltration of articular tissue and the increased levels of pro-inflammatory cytokines in MSU crystals-treated rats. Moreover, doliroside A (DA) and medicagenic acid-3-O- β -D-glucopyranoside (MG) were isolated and identified from EE. The major components of EE, including DA, MG and other triterpenoids, were well confirmed by HPLC. A further study revealed that EE, DA and MG (10, 20, 40 μ g/mL) exhibited stronger inhibitory effects on the production of pro-inflammatory cytokines (including interleukin-1 β , interleukin-6 and tumor necrosis factor- α) in MSU crystals-treated RAW 264.7 cells.

Conclusion: These findings indicate that the major triterpenoids present in DF have a remarkable effect on improving symptoms of acute gouty arthritis induced by MSU crystals through inhibiting the production of pro-inflammatory cytokines.

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1. Introduction

Gouty arthritis, usually characterized as an intense acute inflammatory disease, is provoked by activation of multiple inflammatory mediators by the elevated deposition of needle-like crystals of monosodium urate (MSU) crystals in the joints. This process is driven by neutrophil influx into the joint and leads to the attack of acute inflammatory arthritis with severe pain in the affected tissue. This disease is the most common form of inflammatory

arthritis in men older than 40 years, and affects 1–2% of adults in the developed countries (Krishnan et al., 2008).

It has been reported that MSU crystals stimulated macrophage to release interleukin-1 β (IL-1 β), which is mediated by NLRP3 inflammasome activation (Agostini et al., 2004; Chen et al., 2006). The NLRP3 inflammasome promotes the activation of caspase-1 to cleave the precursor of IL-1 β into its active form (Martinon and Tschopp, 2007). This response is central to the experimental gouty inflammation (Martinon et al., 2006).

Nowadays, a number of anti-gout agents including non-steroidal anti-inflammatory drugs have been available in the market, such as indomethacin and naproxen. They are frequently used as the first-line therapies for acute gout. The associated adverse reactions such as gastrointestinal toxicity, renal toxicity, or gastrointestinal bleeding and so on, however, always limit their clinical uses. Therefore, it is very urgent and important to search for better anti-gouty arthritis drugs. Recently, many herbal drugs and their active ingredients are

Abbreviations: DF, *Dolichos falcata* Klein; DA, doliroside A; EE, ethanol extract; ELISA, enzyme-linked immunosorbent assay; IL-1 β , interleukin-1 β ; IL-6, interleukin-6; MG, medicagenic acid-3-O- β -D-glucopyranoside; MSU, monosodium urate; PBS, phosphate buffered saline; TNF- α , tumor necrosis factor- α .

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increased interest in possible roles in the treatment of gouty arthritis, but the specific mechanisms of action and bioactive chemical constituents of many of these herbs remain unclear.

Dolichos falcata Klein (DF), a Chinese Dai ethnic medicine popularly known as “Tuoyeteng” in Yunnan province of China, has been widely used as a traditional herbal medicine for the treatment of fracture and beriberoid disease for a long time in China (Tian, 2000; Lin et al., 2003; State Administration of Traditional Chinese Medicine, 2005). In some regions, DF is popularly employed in medicinal preparations, such as “Yunnan Hongyao Jiaonang” in the market and “Damayao San” in clinical application (Yu, 2000), for the treatment of fracture, rheumatoid arthritis, and soft tissue injuries by dispelling wind-evil and eliminating dampness, and activating blood flow and removing blood stasis according to the theory of Chinese medicine. Although the previous reports on DF include the isolation of saponins and flavonoids (Pu et al., 1984; Peng et al., 2007; Wang and Wang, 2011) and the anti-tumor and anti-virus activities (Huang et al., 1982), few studies are available on the traditional pharmacological properties of the extract of this plant. In this sense, the aim of the present study was to investigate the components of DF, and to assess the anti-inflammatory activities of the components and the extract of DF, as well as the possible mechanism on gouty arthritis in an animal model of the MSU crystals-induced gouty inflammation.

Anti-gouty arthritis effects of the ethanol extract (EE) of DF were investigated through evaluating the indexes of the pain threshold value, the joint swelling degree, the inflammatory cell infiltration of articular tissue and the levels of pro-inflammatory cytokines in MSU crystals-treated rats. Subsequently, the components of EE were isolated and identified using classical methods. Phyto-chemical analysis of EE was further carried out by HPLC–DAD. Finally, the anti-inflammatory effect of EE and two isolated compounds were assessed using the MSU crystals-treated monocyte/macrophage cell line RAW 264.7 *in vitro*. The final results may support the wide utility of DF in Chinese Dai ethnic medicine.

2. Materials and methods

2.1. Plant materials

The raw materials of the fresh rhizome of DF were collected in July 2011 in Yunnan province of China. The plant materials were identified by Prof. Dingrong Wan, College of Pharmaceutical Sciences, South-Central University for Nationalities, China. The voucher specimen (No.: S20120725) has been deposited at the Herbarium situated in College of Pharmaceutical Sciences. The coarse powder of rhizome was air-dried in the shade and stored in a well-closed vessel for use. The powder (1 kg) were grounded and extracted twice for 2 h each time by refluxing in 60% ethanol (1:10, w/v). After being filtered and centrifuged, the filtrates were concentrated under reduced pressure and 150 g ethanol extract (EE, 15%, w/w) was obtained.

2.2. Reagents and drugs

Analytical grade of chloroform and ethyl acetate were purchased from Nanjing Hanbang Chemical Reagent Company (Nanjing, PR China). Chromatographic grade of acetonitrile purchased from Tedia Company Inc. (USA). Colchicine was bought from Sigma Chemicals (St. Louis, MO, USA). Enzyme-linked immunosorbent assay (ELISA) kits of IL-1 β , tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6) were obtained from R&D Systems Inc. (Minneapolis, MN, USA). EE was prepared by our lab (No.: 20120812). All other reagents used were standard laboratory reagents of analytical grade and were purchased locally.

MSU crystals were prepared by crystallization of a supersaturated solution of uric acid (Aldrich Chemical Company, Inc) under mildly basic conditions. Briefly, 250 mg uric acid was added to 45 mL of ddH₂O containing 300 μ L of 5 M NaOH and the solution boiled until the uric acid was dissolved. The solution was passed through a 0.2 μ m filter and 1 mL of 5 M NaCl was added to the hot solution that was then stored at 26 °C to allow crystallization. After 7 days, the resulting MSU crystals were washed with ethanol and acetone over a Buchner flask and allowed to air dry under sterile conditions. MSU crystals were confirmed as endotoxin-free by a commercial test kit of limulus amoebocyte lysate assay (< 0.01 EU/10 mg).

2.3. Cell and animals

Monocyte/macrophage cell line RAW 264.7, obtained from Cell Bank of the Chinese Academy of Sciences (Shanghai, China), were cultured in DMEM containing 10% heat-inactivated FBS, 25 mM HEPES (pH 7.5), 100 U/mL penicillin and 100 μ g/mL streptomycin. The RAW 264.7 cells were incubated at 37 °C and maintained in a humidified atmosphere containing 5% CO₂.

Sprague–Dawley rats (male and female in half, 230–270 g) were purchased from the Laboratory Animal Center of Huazhong University of Science and Technology (Wuhan, China) and were housed 1 week to adapt to their environment before being used for experiments. All the animals were maintained on standard laboratory conditions of temperature 25 $\pm$ 2 °C and a 12-h light/12-h dark cycle with free access to commercial food and pure water for the duration of the study. Animal experimentation and the corresponding protocol (No.: 2012-0018; Date: 7/15/2012) were approved by the Animal Ethics Committee of South-Central University for Nationalities (Wuhan, China). All the procedures were in strict accordance with the PR China legislation on the use and care of laboratory animals.

2.4. Isolation and identification of compounds from EE

EE (50 g) was subjected to repeated chromatography on silica gel column and eluted with chloroform–methanol (20:1–5:1) gradient solvent system. Further purification was performed by using Sephadex LH-20 chromatography, preparing TLC and other methods. The structure of compound was confirmed by spectroscopic methods (UV, IR, MS, ¹H NMR and ¹³C NMR).

2.5. Analysis of EE by High-performance liquid Chromatography (HPLC)

The multi-components of the EE were characterized by HPLC. The samples were analyzed using Pheccda C₁₈ column (150 mm $\times$ 2.1 mm, 3 μ m) with the detector wavelength set at 215 nm, and the mobile phase consisted of water containing acetonitrile (A) and 0.1% (v/v) acetic acid (B). A gradient program was used as follows: 0–10 min, 32–55% A; 10–20 min, 55–100% A. The flow rate was 0.3 mL/min.

2.6. Rat's experiment

2.6.1. MSU-induced gouty arthritis model and drug administration

A homogenous suspension of EE and colchicine were made with 0.5% carboxy methyl cellulose in phosphate buffered saline (PBS). Fresh solution was prepared before each experiment. All the rats were divided at random into six groups of 12 rats each. According to the previous reports (Lee et al., 2009), rats were anesthetized with 2.5% isoflurane, and then MSU crystals suspension (0.5 mg MSU crystals suspended in 50 μ L endotoxin-free PBS) or PBS were injected into the tibiotarsal joint (ankle) of rats. Group I

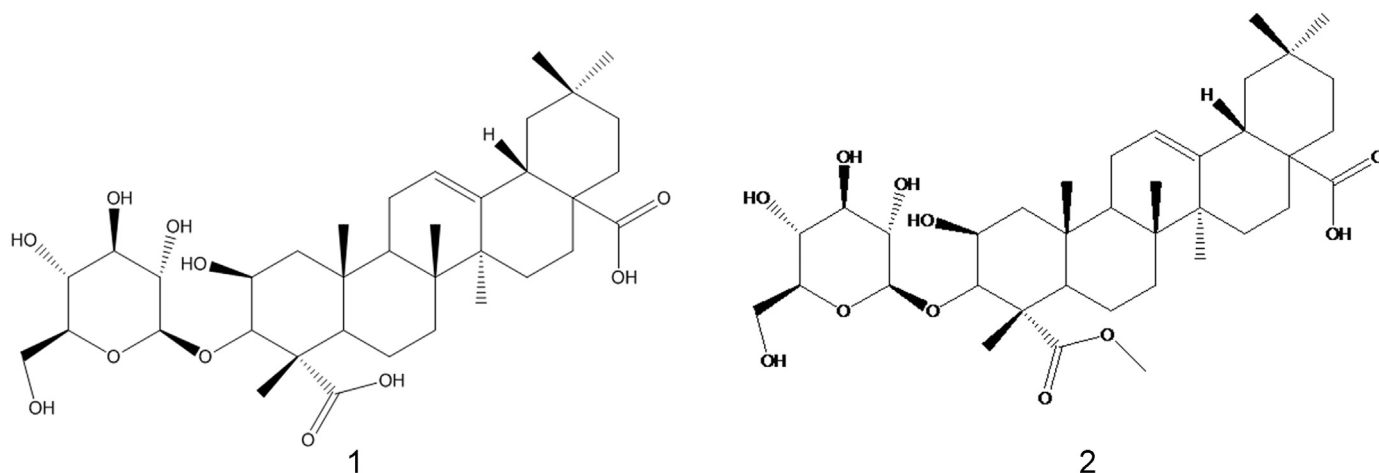


Fig. 1. The chemical structures of medicagenic acid-3-O- β -D-glucopyranoside (1) and doliroside A (2).

injected with PBS served as a normal group. Group II injected with 50 μ L MSU crystals suspension served as a control group. Group III comprised of MSU crystals-treated rats that were administered with colchicine (1 mg/kg body weight). Group IV, Group V and Group VI MSU crystals-treated rats were administered with EE (10, 20 and 40 mg/kg body weight, respectively). EE and colchicine were given orally 1 h after the MSU crystal injection (single dose) and once daily up to 4 days.

2.6.2. Behavioral testing

Before behavioral testing, rats were acclimatized to the testing room for at least 1 h. All behavioral measures were taken by experimenters blind to treatment. Baseline readings were obtained for each measure before injections.

To study thermal hyperalgesia after ankle injections, Hargreaves' test was performed as previously described (Sluka et al., 1997). Briefly, a radiant heat source was applied to the heel and the paw withdrawal latency was recorded. For each rat, three readings were obtained and the mean value was used. The results are presented as the percentage change from the baseline readings.

To test willingness to bear weight on injected ankles, the weight bearing test was performed using an Incapacitance meter (NatureGene Corp., USA). Each hind paw was placed on a transducer pad and the weight distribution, in grams, was recorded over 5 s. The results are presented as percentage weight distributed to the affected limb.

The diameter of ankle joints was measured with calipers while rats were anesthetized by 2.5% isoflurane. The results are presented as percentage change from baseline diameter.

2.6.3. The levels of cytokines in rats' serum and joint

Six rats per group were killed 1 day after ankle injection. Whole blood samples were collected by cardiac puncture from anesthetized rats and allowed to clot for approximately 1 h at room temperature, and then centrifuged at 2500g for 10 min to obtain the serum. The articular cavities were irrigated with 0.5 mL of normal saline. The levels of TNF- α , IL-1 β and IL-6 were measured by ELISA kits (according to manufacturer's instructions) in serum samples and rinse solution of articular cavities.

2.6.4. Histological analyses

For histological analyses, the ankle region was isolated and decalcified in 100 g/L ethylene dinitrilotetraacetic acid, then embedded in paraffin and sectioned at 5 μ m staining with haematoxylin and eosin.

2.7. Cell experiment

2.7.1. Cell viability assay

Cell viability was measured using the MTT bioassay. Briefly, Raw 264.7 cells (1×10^5 cells/well) were seeded in each well of 96-well plate for 4 h. After incubation with various concentrations (10, 20, 40 μ g/mL) of EE, doliroside A (DA) and medicagenic acid-3-O- β -D-glucopyranoside (MG) for 24 h, the MTT solution (5 mg/mL in PBS) was added (10 μ L/well). This plate was incubated for additional 4 h at 37 $^{\circ}$ C, and the formazan crystals produced were dissolved in 100 μ L of DMSO. The microplates were shaken vigorously to ensure complete solubilization. The optical density at 570 nm (OD_{570 nm}) was determined in a Micro-plate Reader (Thermo Multiskan Ascent). All values were presented as means $\pm$ S.E.M. of numbers obtained from six wells and from three separate experiments.

2.7.2. ELISA for cytokine determination

After pre-incubation with varying concentrations (10, 20, 40 μ g/mL) of EE, DA and MG for 20 h, RAW 264.7 cells were treated with MSU crystals (200 μ g/mL) for 4 h. Then, the culture supernatants were collected and used for measuring TNF- α , IL-1 β and IL-6 by ELISA kits. Absorbance was measured in an ELISA reader (Thermo Multiskan Ascent) at 450 nm. The lower limit for detecting cytokine was < 10 pg/mL with intra- and inter-assay coefficient of variations less than 10%. Measurements performed in triplicate. Values represent mean $\pm$ S.E.M. Results are representative of three separate experiments.

2.8. Statistical analyses

All values were expressed as the mean $\pm$ S.E.M. and were analyzed by one-way analysis of variance (ANOVA) followed by the Tukey–Kramer test using the SPSS version 13.0 software; a *p*-value of less than 0.05 was considered significant.

3. Results

3.1. Components isolated from EE

Two components, namely medicagenic acid-3-O- β -D-glucopyranoside (MG, yield 1.2 g) and doliroside A (DA, yield 0.2 g), were isolated and purified from EE. The spectra data were in consistent with the reports in the literatures (Pu et al., 1984). Structures of the

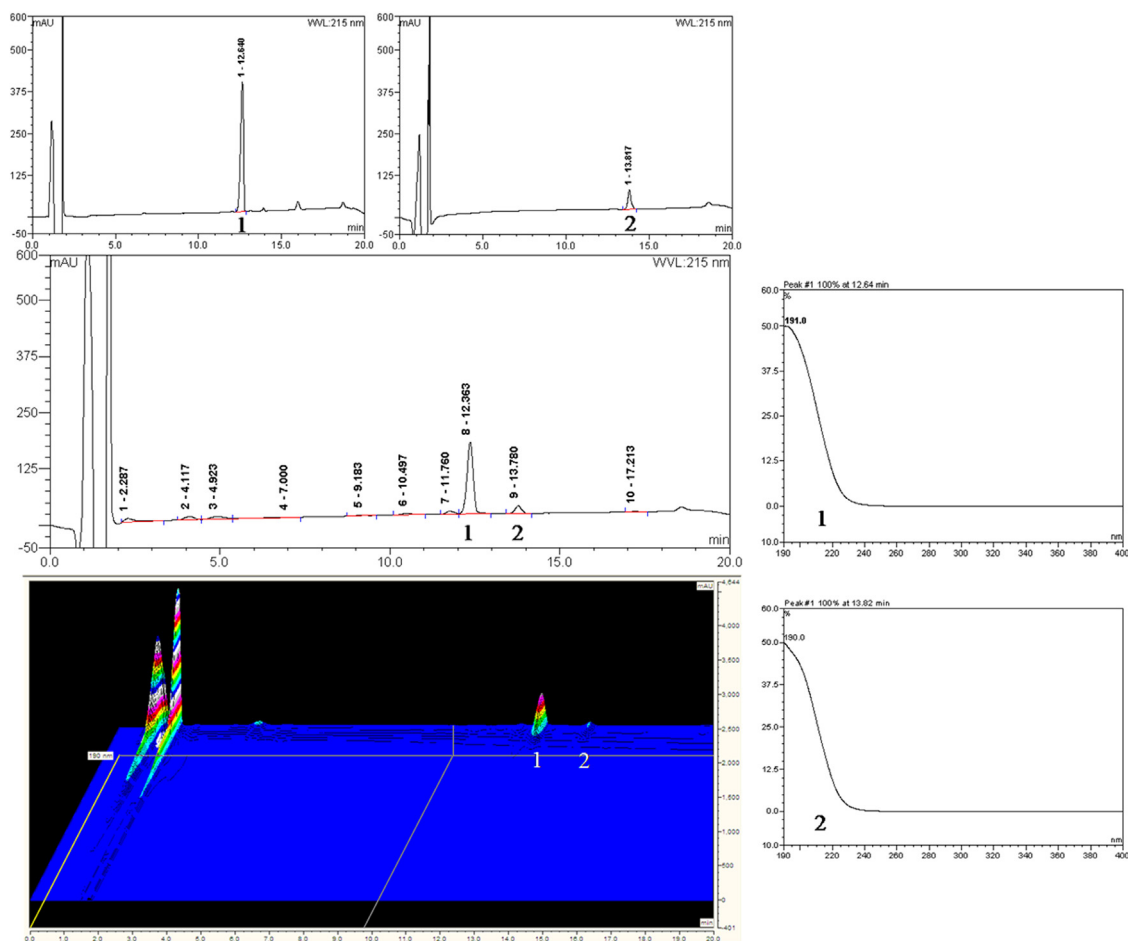


Fig. 2. The typical HPLC-fingerprint of EE. (1) Medicagenic acid-3-O- β -D-glucopyranoside. (2) Doliroside A.

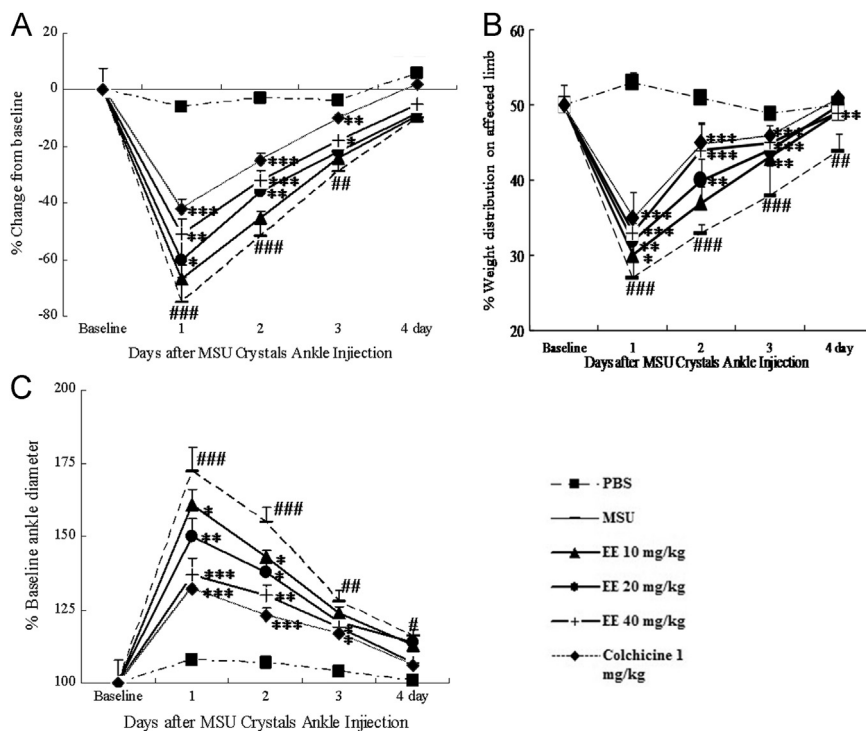


Fig. 3. EE was as efficacious as colchicine in relieving established pain and swelling when given 1 day after ankle injection of MSU crystals ($n=6$). (A) EE and colchicine treatments significantly reduced thermal hyperalgesia for 3 days after treatments. (B) Rats treated with both EE and colchicine showed a significant decrease in the weight-bearing asymmetry. (C) Ankle diameter studies showed that rats treated with EE and colchicine experienced similar reductions in ankle diameter. (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ vs. control group; # $p < 0.05$, ## $p < 0.01$ and ### $p < 0.001$ vs. MSU crystals group, $n=8$). Results are expressed as the mean $\pm$ S.E.M.

two components are shown in Fig. 1. Their purities were all proved to be larger than 95% by HPLC analysis.

3.2. HPLC-fingerprint of EE

The typical HPLC-fingerprint of EE is shown in Fig. 2. By comparing the retention times and the UV spectra of the reference standards, two compounds MG (1) and DA (2) in EE were well identified.

3.3. EE improved the behavior and the joint dysfunction of MSU crystals-treated rats

Animals exhibited thermal hyperalgesia for 3 days after an intra-articular ankle injection of MSU crystals, peaking 1 day after injection. This thermal hyperalgesia was significantly reduced in EE and colchicine groups compared with the control. No hyperalgesia was detected in rats with ankle injection of PBS (Fig. 3A).

Ankle injections of MSU crystals, but not PBS, induced a significant shift in weight bearing away from the affected limb in the normal group, peaking 1 day after injection. In contrast, EE and colchicine groups showed significantly less weight redistribution (Fig. 3B).

Moreover, after injections of MSU crystals, rats developed ankle swelling that peaked after the first day. Both EE and colchicine groups showed significantly less swelling (Fig. 3C).

Histological assessment of joint section from rat's ankle was also conducted. The results demonstrated that EE and colchicine could improve the synovial hyperplasia, reduce the infiltration of

inflammatory cells in the synovium, and diminish the erosive damage in the cartilage (Fig. 4).

3.4. EE attenuates the levels of cytokines in the MSU crystals-treated rats

To identify the cytokines altered by ankle injection of MSU crystals, as well as to study the role of IL-1 in the disease process, we determined the levels of cytokines in the serum and rinse solution of articular cavities collected 24 h after ankle injection. ELISA results confirmed that IL-1 β level was increased in MSU-injected ankles. Moreover, the levels of IL-6 and TNF- α were up-regulated in the serum and the joint of MSU-treated rats. A significant attenuation of MSU crystals-induced increases in the levels of cytokines was observed after the treatments with EE and colchicine (Table 1).

3.5. Cell viability

The cytotoxicity of EE, DA and MG in Raw 264.7 macrophages cells was measured using MTT assay. The cell viability was more or less 96% in all cases. Accordingly, cytotoxic effects of EE, DA and MG on the cell viability were not assessed in the next used concentrations.

3.6. Effects of EE, DA and MG on the secretion of IL-1 β , IL-6 and TNF- α

To confirm the inhibitory effects of EE, DA and MG on the production of inflammatory cytokine in MSU crystals-treated RAW

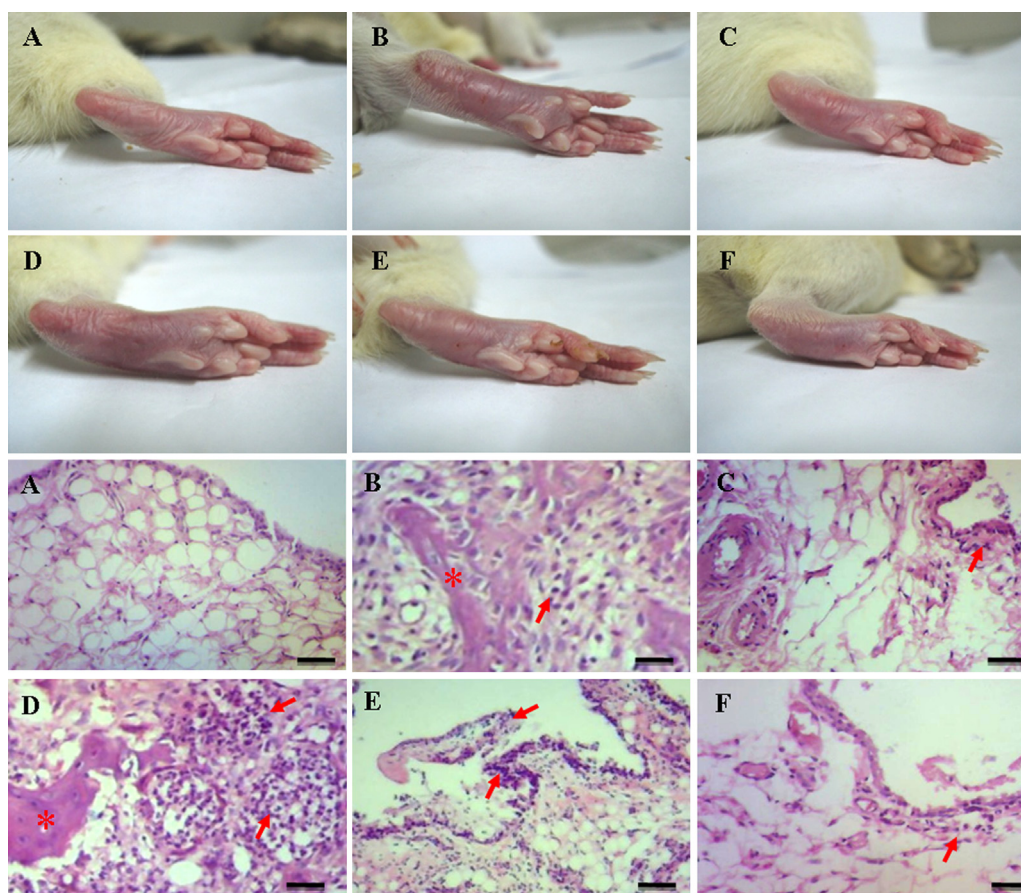


Fig. 4. Inflammatory responses were attenuated by EE and colchicine in MSU crystals-induced acute gout model ($n=6$). (A) Histological analysis of ankle joints from treated or non-treated rats. (A) PBS group; (B) MSU crystals group; (C) colchicine-treated rats; and (D)–(F) 10, 20 and 40 mg/kg EE-treated rats; arrows hinting to inflammatory cells and asterisks hinting to synovial hyperplasia; scale bar=100 μ m; original magnification 200 $\times$.

Table 1
A dose-dependent decrease in TNF- α , IL-1 β and IL-6 was achieved with the increasing doses of EE in the MSU crystals-induced acute gout model. Results are expressed as the mean $\pm$ S.E.M.

Group	Dose (mg/kg)	Serum			Synovial fluid		
		TNF- α	IL-1 β	IL-6	TNF- α	IL-1 β	IL-6
PBS	/	45.16 $\pm$ 4.31	7.54 $\pm$ 1.26	13.25 $\pm$ 2.11	104.12 $\pm$ 9.31	27.32 $\pm$ 3.53	22.25 $\pm$ 0.12
MSU	/	70.82 $\pm$ 9.872;###	10.72 $\pm$ 1.871;##	20.63 $\pm$ 2.262;###	152.84 $\pm$ 10.672;###	45.78 $\pm$ 2.182;###	35.63 $\pm$ 4.202;###
EE	10	65.76 $\pm$ 6.02	9.02 $\pm$ 1.04*	18.59 $\pm$ 2.37	145.76 $\pm$ 12.22	39.24 $\pm$ 2.46*	30.41 $\pm$ 3.34*
	20	62.52 $\pm$ 7.17*	8.87 $\pm$ 1.93*	17.46 $\pm$ 1.86*	138.52 $\pm$ 11.67*	32.82 $\pm$ 3.97**	28.74 $\pm$ 4.18*
	40	58.47 $\pm$ 6.45**	8.23 $\pm$ 1.09**	14.34 $\pm$ 2.05**	125.48 $\pm$ 12.63**	30.33 $\pm$ 4.48**	27.26 $\pm$ 3.65**
Colchicine	1	57.62 $\pm$ 5.58**	8.19 $\pm$ 1.50**	15.47 $\pm$ 2.45**	123.32 $\pm$ 10.58**	29.17 $\pm$ 4.54**	25.47 $\pm$ 3.22**

p < 0.05 vs. control group.

* p < 0.05 vs. MSU crystals group, n = 6.

** p < 0.01 vs. MSU crystals group, n = 6.

*** p < 0.001 vs. MSU crystals group, n = 6.

p < 0.01 vs. control group.

p < 0.001 vs. control group.

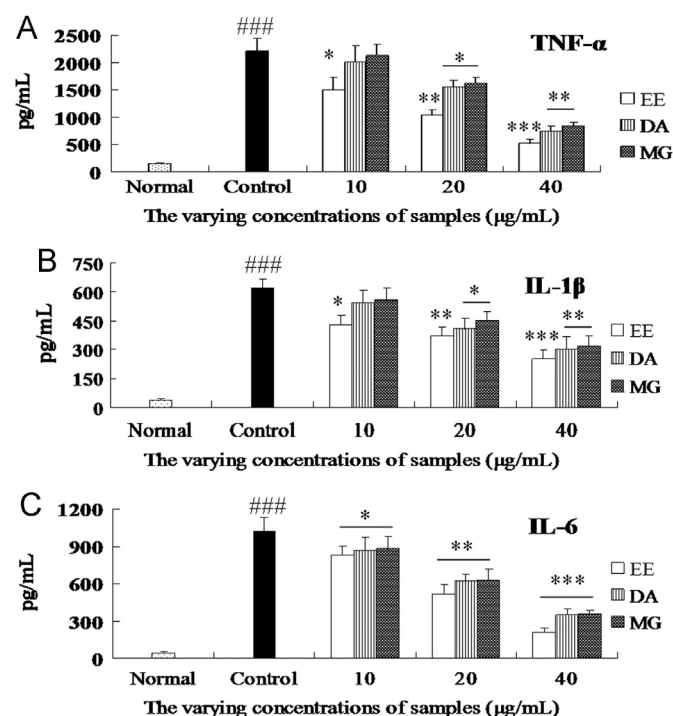


Fig. 5. The inhibitory effect of various concentrations of EE, DA and MG (10, 20, and 40 μ g/mL) on TNF- α , IL-1 β and IL-6 production in MSU treated RAW 264.7 cells by ELISA assay. EE, DA and MG exhibited a suppressive effect on the level of TNF- α , IL-1 β and IL-6 in concentration-dependent manners.

264.7 cells, the levels of IL-1 β , IL-6 and TNF- α were determined by ELISA assay. The results of EE, DA and MG in RAW 264.7 cells at different concentrations (10, 20, 40 μ g/mL) for 24 h are shown in Fig. 5A–C. EE, DA and MG exhibited a suppressive effect on the levels of TNF- α , IL-1 β and IL-6 in the concentration-dependent manners.

4. Discussion

The key features of gouty arthritis are characterized by intense pain, swelling, and reddening around the joints and connective tissues. Recent studies indicate that MSU crystals are among the most potent pro-inflammatory stimuli and an innate immune

inflammatory response to the crystals' surface is intimately involved in the pathology of gouty arthritis (Popa-Nita and Naccache, 2010). The formation of MSU crystals into the joint cavity promotes an acute inflammation, and MSU crystal is most frequently employed to develop an animal model of gouty arthritis (Martin et al., 2009; Torres et al., 2009). In this study, MSU crystals injection obviously induced the joint swelling, elevated the thermal hyperalgesia, and increased the infiltration of inflammatory cells in rats. EE at different doses significantly attenuated the pain threshold value, the joint swelling degree, and the increased inflammatory cells infiltration of articular tissue in MSU crystals-treated rats. This result indicated that EE could ameliorate the gouty arthritis induced by MSU crystals.

Several lines of evidence indicate that MSU crystals stimulate the synthesis and release of IL-1 β (Roberge et al., 1991). Torres et al. (2009) have convincingly showed that IL-1 was a major trigger of joint inflammation in an animal model of gouty arthritis, where they injected MSU crystals directly into the mouse ankle joint. These observations support a critical role of IL-1 β in the mediation of the MSU crystal-induced inflammatory responses and the clinical effectiveness of IL-1 β in the control of the gouty arthritis. Our present study showed that the level of IL-1 β was elevated in the serum and synovial fluid of the ankle joint in MSU crystals-treated rats. However, the secretion of IL-1 β was effectively inhibited by EE, as well as by colchicine. Furthermore, a central role of TNF- α and IL-6 in the experimental model of acute gouty arthritis *in vivo* has also been documented. MSU crystals stimulate the secretion of TNF- α and IL-6 by synovial cells, monocytes-macrophages and neutrophils, which resulted in an acute inflammation (Di Giovine et al., 1987). This concept has been strengthened by the findings of an open study on the effects of anakinra in gout (So et al., 2007). Therefore, the effects of EE on the levels of TNF- α and IL-6 were also evaluated in the MSU crystals-treated rats. Our data indicated that EE significantly reduced the levels of TNF- α and IL-6 in the serum and synovial fluid of the ankle joint in MSU crystals-treated rats, which also contributed to the improvement of inflammation responses induced by MSU crystals.

A vast number of neutrophils influxing into the joints is the hallmark of acute gouty arthritis. Recent evidence implicates that an increase in macrophage density precedes the neutrophils infiltration and macrophages represents an important path-mechanism in the inflammatory processes (Schiltz et al., 2002). Articular resident mononuclear phagocytes respond to MSU crystals by the activation of the inflammatory cascade, resulting in production of large numbers of pro-inflammatory mediators such

as IL-1 β , IL-6 and TNF- α (Guerné et al., 1989; Di Giovine et al., 1991). According to the anti-inflammatory effect evaluation in the MSU crystals-treated rat abovementioned, EE did show obvious inhibitory effects on the activation of pro-inflammatory mediators. In order to clarify the active component, compounds were isolated from EE through various separation methods and then purified. The data of HPLC–DAD demonstrated that saponins were the main components of EE, which was in consistent with the previous research (Pu et al., 1984). Moreover, two major triterpenoid saponin derivatives, namely DA and MG, were well identified. To address whether the anti-inflammatory effect was owing to the components isolated from EE, we investigated the effects of EE, DA and MG on the secretion of pro-inflammatory mediators using RAW 264.7 cells. The results demonstrated that EE, DA and MG (20 and 40 μ g/mL) significantly repressed the production of IL-1 β , IL-6 and TNF- α from RAW 264.7 cells to different degrees in comparison to the only MSU crystals-treated cells. Moreover, EE (10 μ g/mL) significantly reduced the secretion of IL-1 β and TNF- α up to certain level, but lower concentration of DA and MG did not. This result indicates that the anti-inflammatory effect of DA and MG may contribute to the activity of EE on ameliorating the gouty arthritis induced by MSU crystals, and the synergistic interaction of the ingredients in EE may be associated tightly with the effect of EE.

Nowadays, it has long been recognized that the saponins possess significant anti-inflammatory activity and play a significant role in drug discovery and development for the treatment of some inflammatory diseases (Joh and Kim, 2011; Paul et al., 2012). However, the functionality and structure–activity relationships of saponins remain unclear and most researches have exhibited different results. For example, Wang et al. (2008) have showed that CH₂OH at the C-4 position of bayogenin and the carboxyl group at C-17 of the aglycone seem crucial for the acute anti-inflammatory effect in the chemical structure of the saponins. Suh et al. (2007) considered CH₃ at the C-4 position also has a good anti-inflammatory effect. Gao et al. (2009) have reported triterpenoid saponins but not their respective glycosides (including C-17 of the aglycone) were crucial for the anti-inflammatory effect. In the present study, DA and MG with different substituting groups at the C-4 and C-17 also showed significant anti-inflammatory activity. Just as Francis et al. (2002) mentioned, with the information presently available, it is difficult to establish clear functionality and structure–activity relationships regarding the effects of saponins in biological systems.

5. Conclusion

In conclusion, the data from the murine model of acute gouty arthritis and the *in vitro* assays has shown for the first time that DF could ameliorate the gouty arthritis induced by MSU crystals through the inhibition of the synthesis and release of pro-inflammatory cytokines from activated macrophages. DA and MG isolated from DF may be the active components to contribute to the anti-gouty arthritis effect of DF. These findings suggest that the use of DF may provide an effective treatment option for gouty arthritis. Whether DF might exert anti-inflammatory effects by preventing NF- κ B signaling or MAPK pathway is in progress. The effect of DA and MG on the pathogenesis of gouty arthritis in an animal model should also be further investigated.

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